

PROmpter — Manual

PROmpter automatically turns the export files of the Roche **LightCycler PRO** into a readable **PDF report** and an **Excel spreadsheet** (for later LIS processing).

Important: PROmpter is a pure **presentation/reporting tool**. All values (Cq, finalCall, curves) come **unchanged from the LightCycler PRO**. PROmpter does not recalculate anything and does not reinterpret anything — it merely summarises the device's own results clearly. The reports are an **interim evaluation**; the professional sign-off is still done manually. PROmpter is **initially intended for qualitative analyses** — i.e. the **reactive / non-reactive** call per target, as the LightCycler PRO provides it in finalCall. **Quantitative** evaluations (absolute or relative quantification via standard curves) are **currently not** supported.

In a nutshell

1. In the **Settings**, choose a **watched folder** (where the LC PRO drops its export files).
2. **Start monitoring** — done. Every new file is processed automatically.
3. For each file a **PDF** and an **Excel** are created; the **original** is then moved to the archive (or deleted — depending on the setting).

PROmpter may run in the background: with **"Work in background"** it sits as an icon in the notification area (tray) next to the clock, and monitoring stays active. The **window's close button ()**, in contrast, **quits PROmpter completely**.

Installation & first start

PROmpter does **not** need to be installed — it is a single .exe. It is best to copy it to a **fixed local location** (e.g. C:\PROmpter\) and start it by double-clicking; a desktop shortcut makes it convenient. **Administrator rights are not required** — neither on the first nor on later starts.

On the very first start Windows may show a blue message **"Windows protected your PC"** (Microsoft Defender SmartScreen). This is **not** an error and **not** a virus, just a notice because the program is not (yet) digitally signed. Here is how to continue:

1. Click **"More info"**.
2. Then click **"Run anyway"**.

This is only necessary **once per program version** — after that PROmpter starts without asking. (If the file is in a OneDrive folder, the message may reappear; at a local location such as C:\PROmpter\ it stays a one-time thing.)

Note for clinical/laboratory IT

PROmpter is a **stand-alone, portable** Windows application (Python/PySide6/Qt) **without an installer, without administrator rights and without a network service**. It reads only the configured **watched folder** and writes **PDF/Excel** to the chosen output folders, plus its settings to %APPDATA%\PROmpter\. The **only** optional outgoing network access is a **version check** when clicking "" (fetching a small text file; can be disabled by leaving out the update URL).

So that staff do not see the SmartScreen message at all, IT can **whitelist** the file — e.g. via a SmartScreen exception or through AppLocker/WDAC. The cleanest long-term solution is to **sign** the .exe with a **code-signing certificate** of the institution; then the warning disappears entirely.

Exporting the run at the LightCycler PRO

For PROMptter to process a run, it must be exported at the LightCycler PRO as **customer data** (a ZIP file with XML). Here is how at the device:

1. Open the **Results** application and tick the **completed run** in the **run overview**.
2. Choose the **Export** button.
3. In the "**Select location for file and export**" dialog, choose the option "**Export customer data**".

Important: do *not* choose "Perform data export" — that creates an **LCRD** file (folder run_data) for the Roche development software. That file is **AES-encrypted** (Roche's protected internal format) and **cannot** be read by PROMptter at all. PROMptter needs the open **customer data** (ZIP/XML).

4. Choose the **location** (SFTP server or USB device) and confirm with **Export**.

The customer data is stored as a **ZIP archive** (named after the device serial number) in the customer_data folder; inside it there is one XML per run (named after run ID + timestamp). You set **exactly this target folder** (or the folder into which the SFTP transfer lands) as the **watched folder** in PROMptter — then every new export is automatically processed into PDF + Excel.

Tip: Several completed runs can also be **exported together** — PROMptter then creates a separate report for **each run** (see "Several runs in one file").

The main window

- **Start/Stop monitoring** — switches the automatic folder monitoring on/off.
- **Process file now ...** — select and process a single export file by hand.
- **Scan folder now** — search the watched folder once for files not yet processed.
- **Settings ...** — all options (see below).
- **Work in background** — places the window into the notification area (tray); monitoring keeps running in the background (bring it back via the tray icon's "Show", "Quit" closes it completely).
- **Close** — the **window's close button** () quits PROMptter **completely** (not to the notification area).
- **Log** — shows for each file the time, name, status (**green** = OK, **yellow** = notice/partially processed, **red** = error) and details. If something fails, **the original file stays put** — nothing is lost.
- **Save log ...** — save the displayed lines as **CSV** (for Excel) or as a **text file**.
- **Clear log** — clears only the display; the PDF/Excel files already created are kept.

Settings

The settings are organised in four tabs: **File management** (folders, which files, original action, printing, file names), **PDF report** (sections, header/footer, sorting, curve type), **Excel report** (columns, sorting, sheets, curve type) and **CSV/TXT export** (Roche-style format, columns, delimiter). **Sorting and curve type can be set separately per report** — the PDF may e.g. be by column and the Excel by row.

For **sorting the samples** there are four options: **by column** (A1, B1, C1 ...), **by row** (A1, A2, A3 ...) as well as **alphabetically ascending or descending** by the **sample name**.

Sprache / Language

At the very top of the **File management** tab, the **language** can be changed: **Deutsch** (default) or **English**. The change affects the **interface, the PDF report, the Excel file and all messages**; in English mode numbers are written with a **point** instead of a comma (e.g. Cq 15. 62). **Device and sample data** (assay/sample names, Cq, finalCall, curves) stay unchanged in **both** languages. The new language takes effect **after a restart** of the program. (*The manual is available in German and English and opens automatically in the language that is set.*)

Folders

- **Watched folder** — is monitored automatically.
- **PDF output / Excel output** — target folders. **Empty = same folder as the export file** (the finished PDF/Excel are not read in again).
- **Archive folder** — where the original is moved to. Empty = subfolder Archiv/. (Deliberately a **separate** folder so that the moved original is not immediately processed again as a "new file".)

Original file after processing

- **Move to archive** (default) or **delete**. *"Leave in place" deliberately does not exist*, so that the same run is not accidentally processed twice.

Printing

PROMptter can **automatically print the PDF report after it has been created** — handy for the paper archive of an unattended watched folder.

- The **checkbox "Print PDF report automatically after creation"** turns the function on (default: off).
- **Printer** — the **default printer** (Windows) or a specific printer from the list of installed printers. This lets the monitoring PC print specifically to the **lab-report printer**.
- **Copies** — number of printouts per report (e.g. 2 for a duplicate archive).
- **PROMptter prints the PDF itself** directly to the chosen printer — **regardless of which PDF program** is installed or set as default on the PC (no Adobe/Edge needed).
- **Safety: only the generated PDF** is ever printed, never the original file. If printing fails once (e.g. no printer reachable), the run is still **counted as processed** (PDF/Excel are saved after all) and marked **yellow** with a notice in the log.

Sorting the samples

- **by column** (A1, B1, C1 ...) or **by row** (A1, A2, A3 ...) — selectable **separately** in the *PDF report* and *Excel report* tabs.
- **Controls (NTC/PTC)** always appear in the Excel, clearly marked as a control. In the **summary** (Excel) and the **Cq table** (PDF) they are additionally highlighted: **PTC red, NTC green, reactive samples bold**.

Curve display

Which curve form computed by the device is displayed (selectable **separately** in the *PDF report* and *Excel report* tabs). In brackets the name that the **LightCycler PRO** itself uses for it:

Selection (PROMptter)	At the device	Meaning
Baseline divided (default)	"Baseline divided" (bdcurve)	Each curve point is divided by the baseline . This is what the LC PRO always computes its results from — hence the most meaningful display.
Baseline subtracted	"Baseline subtracted" (bscurve)	The baseline is subtracted from each point .
Raw values	"Raw data" / "Fluorescence history"	The unprocessed measured fluorescence.

The **baseline** is the y-axis intercept of an amplification curve; each curve has its own. All three forms come **unchanged from the device** — PROMptter does not compute its own baseline. (*Terms as per the LightCycler PRO user assistance.*)

Two curve views (each individually switchable on/off in PDF and Excel):

- **per sample** — a small chart per well with all channels (overview per patient sample); the targets/channels are coloured as a **spectrum blue -> dark red**.
- **per target** — *one* chart per channel/target, in which **all samples lie on top of each other** (the classic amplification-curve view). The curves are **coloured by result** (reactive coloured, negative grey, controls highlighted); reactive curves and controls are **numbered**, with a small **mapping table** next to them (no. -> well -> sample -> Cq). In Excel, each sample is additionally its own line whose name appears **on mouse-over**. The **Y scale is aligned per target to the strongest sample** (the fluorescence level differs from target to target).

File names of the reports

The name of the generated PDF/Excel is assembled from **building blocks**. Buttons insert placeholders; the **preview** immediately shows the result. *(The placeholders are inserted by buttons and are the same in both languages.)*

Placeholder	Content
{Plattenname}	Name of the plate setup, e.g. "260609_BovPev_Myco"
{Datum}	Run date, e.g. 2026-03-27
{DatumZeit}	Date + time
{Uhrzeit}	Time
{LaufID}	Run ID
{PlateID}	Plate ID
{LCAP}	1st LCAP of the run, e.g. "GastroV96" (the old name {Assay} still works)
{Profil}	PCR profile, e.g. "mPCR7"
{LCAPVersion}	Version of the 1st LCAP, e.g. "1.0" (the old name {AssayVersion} still works)
{Geraet}	Device serial number
{Bearbeiter}	Operator from the file

Free text is allowed (e.g. Result_{Datum}_{LaufID}). Empty blocks and invalid characters are cleaned up automatically; if the name is the same, PROmptter appends _2, _3 ... and **never overwrites**.

Header and footer of the PDF

- **Header** — up to **two freely chosen lines** (e.g. "Example University" / "Virus diagnostics"). Below it the **plate name - run ID** is shown.
- **Footer (left)** — freely composed from **building blocks** (the same placeholders as for the file name), default {Datum}_{PlateID}_{LaufID}. The **page number** stays on the right.

PDF — which sections appear

Header, control-validity check, plate overview, result overview, individual values (Cq), **amplification curves per sample** and **amplification curves per target** — each section individually switchable on/off.

Excel — which sheets appear

As with the PDF sections, a checkbox can be set per **sheet**: **Result table**, **Summary**, **Amplification curves per sample**, **Amplification curves per target**, **Reactive samples**.

Excel — which columns appear

A checkbox per column (for the "Results" sheet). The table is built "long" (one row per sample × target) — good for the later LIS import.

The outputs

PDF report

Header with run key data · control-validity check (NTC/PTC) · coloured **plate overview (96 or 384 wells, automatically detected)** · **result overview per sample** (overall statement, reactive targets with Cq, internal control) · **individual values** (Cq per target) · **amplification curves per sample** (one chart per well, all channels) · **amplification curves per target** (all samples per channel overlaid, coloured by result + mapping table). Uniform Y scale **from 0**, aligned to the samples.

Excel workbook

- **Results** — the configured columns (flags in plain text). The **"Flags"** column shows **all** flags, the separately selectable **"Critical flags"** column only the heavily weighted ones.
- **Summary** — cross table: one row per sample, one column per detection target with the Cq (**empty = not tested**, **"-" = tested but negative**), plus internal control, overall result and the **"Critical flags"** column (plain-text reason), plus a tally.
- **Curves** — native, editable Excel line charts per sample.
- **Curves per target** — *one* chart per channel with all samples (coloured by result; sample name on mouse-over).
- **Reactive samples** — the same per-sample charts, only for reactive samples.

CSV/TXT result table (optional, Roche-style)

In addition to PDF and Excel, PROmpter can write a **CSV/TXT result table** per run — handy for colleagues or LIS tools that expect the familiar Roche tables. Enable it in the *File management* tab (checkbox **"CSV/TXT file"**); the format and options are in a dedicated **"CSV/TXT export"** tab. There are **four formats**:

- **LightCycler 96** (default) — one row per well × target (columns *Position · Sample Name · Gene Name · Cq · Call · Sample Type · Dye · EPF ...*), tab-separated, point decimal.
- **LightCycler 480 – classic** — one block per detection target with a title line (*Include · Color · Pos · Name · Cp · ...*), comma decimal.
- **LightCycler 480 – extended** — text columns in quotes (*Experiment Name · Analysis Name · Position · SampleName · CrossingPoint · ... · Call*).
- **Do-it-yourself** — freely choose columns, delimiter (tab/comma/semicolon), decimal separator, quoting, header row and an optional title line.

Important: Cq and the result (Call) come unchanged from the device. PROmpter is qualitative — so the **concentration and standard-curve columns** of the LC480 formats stay **empty or "-"** (the column layout still stays Roche-compatible).

How PROmpter summarises (transparently)

- **reactive** — as soon as **at least one** target is reported by the device as reactive/positive (the affected targets are named).
- **non-reactive** — all detection targets are non-reactive/negative.
- **see individual values** — mixed or unexpected (e.g. an "Invalid").
- **Internal control (IC)** — is only **shown**; it does **not** automatically block the result.
- **Flags** are shown in **plain text** (e.g. "EPF value too low").

This overall statement is **not a new assessment**, only a bundling of the device's own individual results.

Several runs in one file

If the LightCycler PRO exports **several runs into one file** (one ZIP archive, named after the device serial number), PROMptter creates a separate PDF and a separate Excel for **each run** (named after the respective run). This works **reliably** — so you can safely export several completed runs **together**.

Background: each run is a **self-contained, complete XML file** in the ZIP (named after run ID + timestamp); PROMptter reads them one after another. Should a single run exceptionally be **unreadable** once (e.g. a corrupted export file), PROMptter still processes the remaining runs and marks the file **yellow** in the log with a notice — the whole file is never discarded, and the original is, in this case, **archived instead of deleted** to be safe.

Notes & limitations

- PROMptter is initially designed for **qualitative analyses** (reactive / non-reactive per target). **Quantitative** evaluations (quantification via standard curves) are **currently not** supported.
 - PROMptter is a **helper tool** and **not validated**. Please check the result, controls and plausibility before every sign-off.
 - The export files carry `sentToHost=false` / status "unreviewed" — so they are **not yet released**.
 - PROMptter **never** writes to the original data; it only reads and creates new files.
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License & open-source libraries

- You **may** pass PROMptter on to colleagues and use it in the lab. Private/free distribution is unproblematic.
- PROMptter uses open-source libraries (no part of Roche's own):
- **PySide6/Qt** (© The Qt Company) — license **GNU LGPL v3** · <<https://www.qt.io/licensing>>
- **ReportLab** (PDF generation) — **BSD license** · <<https://www.reportlab.com>>
- **openpyxl** (Excel generation) — **MIT license** · <<https://openpyxl.readthedocs.io>>
- **watchdog** (folder monitoring) — **Apache 2.0 license** · <<https://github.com/gorakhhargosh/watchdog>>

These licenses permit the use in PROMptter without disclosing your own source code. As long as the app is passed on **as an unchanged, stand-alone** .exe, all license obligations are met. (You will find the same note in the **/About** dialog of the app.)

- **Roche** and **LightCycler** are trademarks of their respective owners. PROMptter is an **independent** tool, **not** reviewed or supported by Roche.
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Feedback

Suggestions for improvement and bug reports are welcome — via the **button** in PROMptter (e-mail or online form).